INFRAPULSE

Rural Infrastructure Reliability Intelligence Platform

FULL PROBLEM STATEMENT, EXISTING-SOLUTION REVIEW AND PROPOSED SOLUTION

TechSprint 2026 | 24-hour national-level SDG hackathon

Primary alignment: SDG 9 - Industry, Innovation and Infrastructure

Core proposition
A village can have roads, electricity, broadband and public digital assets recorded as available, while residents still experience unreliable service. InfraPulse measures the gap between infrastructure presence and dependable service, then helps decision-makers prioritize improvements.

Prepared: 9 October 2026
Basis: uploaded InfraPulse concept + public-source research
Status: hackathon concept and MVP specification; not an official government standard

Availability is a starting point. Reliability is the outcome that people experience.

Document Guide

This document retains the original InfraPulse concept and expands it into a research-backed problem/solution proposal. Public-source claims are cited using numbered references in the text and a clickable reference list at the end.

1. **Executive summary** - The proposal in brief
2. **Project definition and scope** - What InfraPulse is and is not
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Research boundary: *This is a public-source review, not a formal procurement, patent, or exhaustive product audit. A feature described as a design opportunity means that the cited public documentation does not establish the exact unified capability proposed here; it does not prove that no agency has a private or newer implementation.*

1. Executive Summary

InfraPulse is a GIS-based Rural Infrastructure Reliability Intelligence Platform proposed for SDG 9. It tackles a specific planning blind spot: the presence of an infrastructure asset is not the same as the reliable delivery of the service that asset is meant to provide.

Official Indian systems already collect village-level development indicators, map rural assets, publish road connectivity data, and support telecom-quality measurement. Mission Antyodaya surveys village/Gram Panchayat indicators and produces rankings and gap reports; Gram Manchitra provides spatial assets and proximity tools; PMGSY has made extensive rural road/facility GIS data public; BharatNet tracks service-ready Gram Panchayats; and TRAI MySpeed measures aspects of real-world network performance. [2]-[9]

InfraPulse is therefore not proposed as another static village map or a replacement for any of those systems. It is a complementary decision layer that combines permitted baseline data with service-performance observations, exposes the availability-reliability gap, estimates who is affected, ranks possible interventions, and lets planners compare scenarios before choosing a priority.

The hackathon MVP should demonstrate five things: (1) an interactive village map; (2) a transparent multi-dimensional reliability score; (3) a gap detector; (4) an explainable priority ranking; and (5) a what-if intervention simulator. An honest, explainable scoring model is more appropriate than an unvalidated claim of high-accuracy AI.

Project thesis: Move rural infrastructure planning from “What assets exist?” to “Which services reliably work, who remains underserved, and which improvement should be prioritized first?”

2. Project Definition, Scope and Boundaries

Project name: InfraPulse
Full name: Rural Infrastructure Reliability Intelligence Platform
Primary SDG: SDG 9 - Industry, Innovation and Infrastructure
Primary users: district planners, Gram Panchayat teams, rural development agencies, infrastructure departments, utilities and civil-society organizations.

2.1 Key terms

Term	Meaning in InfraPulse
Availability	Whether an asset, connection, or service is recorded as present or offered at a location.
Reliability	How consistently the service meets a defined performance or uptime expectation over a stated period.
Accessibility	How far, how long, or how costly it is for people to reach and use the service.
Resilience	How well a service continues or recovers after a disruption; included in future extensions where data support it.
Reliability gap	A difference between the availability score and measured/proxy service-reliability score, when both are comparable.
Priority score	A transparent decision-support score combining service deficit, population impact, criticality, vulnerability and lack of alternatives.

2.2 Intended scope

- Start with a defined pilot geography (one block/mandal or a selected group of villages). Avoid claiming national coverage during the hackathon.
- Measure five dimensions: digital connectivity, electricity reliability, road accessibility, public digital infrastructure, and access to essential services.
- Accept mixed evidence: public datasets, official reports, approved feeds, field measurements and clearly labelled prototype data.
- Show source date, confidence and missing data for every score so a viewer can distinguish verified observations from estimates.

2.3 Outside the initial scope

- Replacing government databases, utility control systems or official planning processes.

- Declaring any village officially “good” or “bad” based on one prototype score.
- Making investment or engineering approvals without field verification and authorized decision-makers.
- Claiming live integrations, predictive accuracy or actual impact unless these have been implemented and validated.

3. Complete Problem Statement

Problem title: Rural infrastructure is measured by presence, while lived service reliability remains hard to compare.

In rural communities, roads, electricity connections, broadband, public digital access points, schools, healthcare facilities and other public assets may appear in administrative records. However, the existence of an asset does not guarantee that residents can depend on it when they need it. Broadband may be deployed but slow or intermittent; a road may be mapped but become impassable in certain seasons; an electricity connection may exist but experience repeated interruptions; a Common Service Centre may be present but have limited operating capacity or poor connectivity; and an essential service may exist nearby in straight-line distance but take a long time to reach over the available road network.

The information needed to understand these issues is often distributed across datasets with different owners, geographic units, definitions, update cycles and levels of granularity. One system may list assets, another may record network tests, another may document road conditions, and another may contain population estimates. Comparing them is difficult without harmonizing village identifiers, geography, dates, units, confidence and the meaning of each indicator.

As a result, a decision-maker may know that infrastructure has been built or a service has been declared ready, but still lack a clear, comparable answer to four practical questions: (1) where is service reliability lowest; (2) which population is affected; (3) which failures most reduce access to essential services; and (4) which feasible intervention would produce the greatest improvement for the available budget?

InfraPulse proposes to close this decision-support gap with a village-level, geospatial reliability profile and intervention-ranking framework. It combines availability with service performance and accessibility, makes uncertainty visible, and translates gaps into a prioritized list of possible actions. The project complements existing systems by using their data where accessible; it does not assume that those systems have no monitoring, analytics, or gap-reporting capabilities.

Research question

Which rural communities have infrastructure that is present but not reliably usable, what evidence explains the gap, who is most affected, and which intervention is most likely to improve reliable access?

3.1 Root causes

- **Presence-oriented indicators:** infrastructure registers often need to record whether an asset or connection exists, but that field alone cannot describe quality over time.
- **Sector silos:** telecom, electricity, roads and public-service data are maintained in different systems and may not share a common village identifier or reporting cycle.
- **Temporal mismatch:** infrastructure location data may be relatively stable, while outages, road closures, network quality and service hours change daily or seasonally.
- **Access is more than distance:** physical barriers, poor roads, service hours, network reliability and affordability affect whether a facility is practically usable.
- **Uneven data coverage:** high-frequency measurements may exist for some regions or users but not all villages; missing data can make comparisons misleading.
- **Weak connection between diagnosis and action:** a gap report can identify a problem but may not compare intervention options, affected population and expected benefit on a common basis.

3.2 Consequences

- Investment choices may overvalue places where infrastructure exists on paper and undervalue communities with unreliable service.

- Students can lose practical access to digital learning; residents may struggle with online services; small businesses may experience interruptions in payments and operations.
- Power disruption can reduce the usability of telecom sites, schools, clinics and digital service points, creating cross-sector effects.
- Poor or seasonal road access can increase travel time and isolate households from healthcare, education, markets and emergency services.
- Departments may duplicate surveys or make separate plans without a shared picture of combined impact.

4. Why the Problem Matters and Who Is Affected

SDG 9 Target 9.1 calls for quality, reliable, sustainable and resilient infrastructure, with affordable and equitable access. The UN indicator framework includes the Rural Access Index (9.1.1), which measures the proportion of rural residents living within 2 km of an all-season road. This establishes both the importance of reliability/access and a useful precedent for transparent geospatial measures. [1][10]

India has also built major digital and geospatial infrastructure foundations. A Government of India release in August 2026 reported about 2.21 lakh Gram Panchayats made service-ready under BharatNet as of June 2026, alongside 3.91 lakh functional rural Gram Panchayat-level Common Service Centres as of May 2026. Those figures are evidence of substantial infrastructure and access-point deployment; they are not, by themselves, direct measurements of a resident's end-to-end service reliability at a particular place and time. [7]

The PMGSY rural connectivity GIS data release reported coverage including over 800,000 rural facilities, 1 million habitations and 2.5 million km of rural roads. Such data is a major foundation for analysis, and InfraPulse should reuse it where the current data, licensing and technical access permit. [5]

4.1 Stakeholders and effects

Affected group	Potential impact of unreliable infrastructure	How InfraPulse supports decisions
Rural households	Unreliable power, connectivity, transport or service access affects daily activities and access to government services.	Shows combined infrastructure gaps and travel/access barriers.
Students and teachers	Poor connectivity, electricity interruption or difficult travel can limit learning and school access.	Highlights villages where school access and digital reliability are weak.
Patients, caregivers and health workers	Travel barriers or unstable electricity/connectivity can make health services harder to reach or use.	Elevates critical-service access and identifies high-impact bottlenecks.
Farmers and rural businesses	Disrupted transport, connectivity or power can affect market access, digital payments and operations.	Combines service reliability with population/economic exposure where data is available.
Women, older adults and people with disabilities	Longer journeys, weak transport and inaccessible service points can impose disproportionate burdens.	Uses vulnerability and service criticality as explicit priority factors where ethically available.
Gram Panchayats and district planners	Fragmented records make comparison, planning and justification of investment difficult.	Provides a common evidence view and explainable ranking.
Utilities and infrastructure departments	Need to locate persistent service-quality issues and coordinate maintenance/upgrades.	Shows dimension-level gaps and possible intervention scenarios.
NGOs and development agencies	Need to identify underserved locations and target limited programme resources.	Provides a documented, repeatable way to compare service deficits.

4.2 Equity principle

A single low score should not automatically label an entire village. InfraPulse should show which service is weak, how confident the evidence is, and which subgroups or facilities may be affected where disaggregated data is lawful and reliable. Rankings should be used to investigate and prioritize, not to stigmatize communities or make eligibility decisions automatically.

5. Existing Solutions: What Is Available and Where the Design Gap Remains

The following review is based on public-facing official descriptions and public project repositories checked on 9 October 2026. It is not an exhaustive audit of private dashboards or non-public government systems. The limitations below therefore describe the gap between each system's documented primary focus and the specific combined workflow proposed for InfraPulse, not proof that a system has no additional capabilities.

5.1 Mission Antyodaya - survey, ranking and gap reports

Mission Antyodaya collects multi-sector village/Gram Panchayat indicators and uses them to create rankings and gap reports that support Gram Panchayat Development Plan (GPDP) planning. The 2022-23 FAQ describes 216 indicators across 21 sections, including infrastructure, roads and communication, electricity, financial/communication infrastructure, health, education and other development subjects. [2][3]

Strength: It provides a nationally significant baseline and helps identify development gaps across sectors. InfraPulse should treat it as a key potential baseline, not duplicate it.

Gap relevant to InfraPulse: survey indicators and composite rankings do not automatically supply a continuous, high-frequency feed of operational performance for every service. InfraPulse would add the service-performance evidence available for the pilot, indicate data age/confidence, and connect the profile to intervention simulation. This is a complement, not a claim that Mission Antyodaya is incapable of all service-level analysis.

5.2 Gram Manchitra - Panchayat GIS and spatial planning

Gram Manchitra provides Panchayat profiles and spatial context for assets including educational institutions, health, financial institutions, PDS and Common Service Centres. Its public interface also lists Mission Antyodaya reports, GPDP layers, road construction/streetlights, nearest-feature and service-area tools. [4]

Strength: It is a close functional neighbour for mapping village assets and analysing proximity. It makes clear that a static map is not sufficient differentiation.

Gap relevant to InfraPulse: the proposed contribution is a normalized multi-sector reliability view that joins the asset's presence to dated service-performance observations and turns the combined view into a transparent priority score and what-if comparison. Any overlapping features must be acknowledged and the project should focus on operational reliability rather than remapping existing assets.

5.3 PMGSY / GeoSadak and rural-road GIS

The Ministry of Rural Development announced open rural connectivity GIS data in 2022, covering 800,000+ rural facilities, 1 million+ habitations and 2.5 million km of roads. The release points to the GeoSadak open-data portal and mentions other road programme systems used for sanction, execution, maintenance and complaint redressal. NRIDA's road master-planning material describes spatial, attribute and operational data, including road networks, terrain, census details, road-condition surveys, health/education/market facilities and traffic data. [5][6][16]

Strength: Excellent potential source for road-network and facility location data.

Gap relevant to InfraPulse: a road-focused layer cannot alone establish whether the village's electricity, broadband and public-service network is reliable. InfraPulse should ingest road data for access analysis and combine it with other permitted sector measurements.

5.4 BharatNet and Common Service Centres - availability / digital access

BharatNet is designed to expand rural broadband infrastructure. The official programme describes FTTH and rural connectivity services. In August 2026, PIB reported about 2.21 lakh Gram Panchayats service-ready under BharatNet as of June 2026 and 3.91 lakh functional CSCs at Gram Panchayat level as of May 2026. [7][8]

Strength: These programmes establish important infrastructure and last-mile service channels.

Gap relevant to InfraPulse: service-ready status or the presence of a CSC is not identical to measured end-user service quality, availability by hour, transaction success, or power/network reliability at a given village. The project should keep deployment indicators separate from observed performance indicators.

5.5 TRAI MySpeed - actual telecommunications quality measurements

TRAI announced a refreshed MySpeed app in February 2026. Its stated tests include download/upload speed, latency, jitter, packet loss, video/web tests, continuous and scheduled tests, and location-based historical performance maps. It says consumer-initiated measurement data is anonymized. [9]

Strength: This is a direct precedent for measuring real-world connectivity rather than only coverage, and is a potential reference for InfraPulse's digital dimension.

Gap relevant to InfraPulse: telecom measurement is one sector. Public documentation does not establish that MySpeed provides the full cross-sector village profile, electricity/road/service accessibility fusion, cost-aware intervention ranking, or multi-sector what-if workflow proposed here. Access to bulk data or an API should not be assumed; the MVP should use authorized exports, permitted public data or team-generated test observations.

5.6 World Bank Rural Access Index - standardized road-access metric

The World Bank's Rural Access Index measures the share of rural population living within 2 km of an all-season road and is SDG indicator 9.1.1. The methodology combines population distribution, road location and road passability, and documentation warns that data quality and update frequency affect results. [10][11]

Strength: It offers a transparent, interpretable geospatial indicator and a strong model for defining a measurable indicator.

Gap relevant to InfraPulse: RAI is deliberately a road-access indicator, not a complete score of electricity, digital connectivity, public digital facilities and service reliability. InfraPulse should preserve the precision and interpretability of such component indicators rather than disguising all differences in one opaque number.

5.7 Related public prototypes and hackathon projects

Several public projects further show that rural GIS, infrastructure scoring and digital twins are not new in isolation:

- **ACCESSLab Rural Infrastructure Assessment:** its repository describes route-level readiness scores combining lane-marking quality, GPS quality, mobile broadband and HD-map availability, with heatmaps and dashboards. This is close prior art for multi-signal infrastructure readiness, but its documented unit is a route/vehicle-readiness assessment. [12]
- **Hills to Highways:** a Hack-a-Sol 2024 track winner describing GIS/ML to identify where roads in rural/hilly terrain should be created or improved, using population, traffic, terrain and economic factors. Its main focus is road planning. [13]
- **Village Digital Twin / RuraLens:** a public repository describes map-based rural asset visualization and a graph-based failure/cascading-impact simulator. It is a community prototype and should not be treated as an independently validated public-sector deployment. [14]
- **SmartGram:** a public repository describes a rural digital twin, GIS, infrastructure dashboard and population/resource forecasting. It establishes prior art for rural digital twins and forecasts. [15]

5.8 Summary comparison

System / project	Documented primary focus	Overlap with InfraPulse	Design opportunity for InfraPulse
Mission Antyodaya [2][3]	Multi-sector survey, composite indicators, rankings and gap reports.	Very high for village data and gap analysis.	Add service-performance time series where available, explicit confidence and scenario comparison.
Gram Manchitra [4]	Panchayat GIS, assets, maps, proximity and service-area tools.	High for geospatial interface and asset discovery.	Focus on reliability profile and intervention logic, not another asset map.
PMGSY / GeoSadak [5][6][16]	Rural roads, habitations, facilities, planning/maintenance data.	High for road/facility geodata.	Join road accessibility with electricity, connectivity and service usage indicators.
BharatNet / CSCs [7][8]	Broadband infrastructure and digital service touchpoints.	High for the digital availability layer.	Separate “service-ready/point exists” from actual usability and uptime.
TRAI MySpeed [9]	Real-world telecom performance testing and geospatial results.	High for the connectivity-reliability component.	Use as one possible signal in a multi-sector decision system, subject to data access.
Rural Access Index [10][11]	Population within 2 km of an all-season road.	High for road-access methodology.	Extend the concept to other services without hiding component-level scores.
ACCESSLab [12]	Route-level road/vehicle/digital readiness.	High for multi-signal readiness scoring.	Differentiate by village-level, multi-sector service access and investment scenarios.
Hills to Highways [13]	GIS/ML for road investment planning.	High for geospatial prioritization.	Prioritize reliability failures and compare repair options, not only new/upgrade road need.
RuraLens / SmartGram [14][15]	Village digital twins, asset models, simulation or forecasting.	High for digital-twin and simulation concept.	Keep scope narrower, use traceable real/proxy data, make uncertainty and intervention economics explicit.

Research conclusion: InfraPulse must not claim that no similar system exists. Its defensible value proposition is the proposed combination of (a) availability-versus-performance comparison, (b) multi-sector reliability and access profile, (c) data freshness/confidence, (d) population- and criticality-aware ranking, and (e) transparent intervention comparison in a single hackathon workflow.

6. The Specific Gap InfraPulse Will Address

Decision question	Typical evidence source	InfraPulse contribution
Does the asset exist?	Asset inventories, programme dashboards, village surveys.	Ingest it as availability baseline, preserve source/date.
Does it work reliably?	Sector tools, network tests, outage/condition logs, field reports.	Convert compatible observations into dimension-level reliability indicators.
What does the resident experience?	Travel time, measured signal/service quality, hours/closures, verified reports.	Add accessibility and usability measures next to asset presence.
How confident is this score?	Often represented differently by source.	Show data coverage, sample count, age and confidence alongside the score.
Who is most affected?	Population, facilities and vulnerability data across sources.	Combine service deficit with people affected and criticality.
What should be fixed first?	Departmental plans, budgets and expert judgement.	Compare transparent intervention scenarios, expected score improvement and cost.

InfraPulse is best understood as an evidence-fusion and decision-support layer. It should integrate data rather than duplicate source-of-truth systems. Each fact displayed must retain its source, observation date, geographic scope and whether it is measured, reported, inferred or simulated.

7. Proposed Solution - InfraPulse

InfraPulse will create a digital infrastructure profile for each pilot village. The profile combines five dimensions and displays both a composite score and the underlying component scores. It then identifies the difference between infrastructure availability and observed performance, estimates service accessibility, ranks high-priority gaps and presents candidate interventions for human review.

7.1 Product modules

Module	What it does	Output
1. Data intake and harmonization	Imports permitted CSV/GeoJSON/table data; maps IDs and geography; standardizes units and dates.	Canonical village and asset records.
2. Data quality and provenance	Checks missingness, invalid ranges, age, sample count, source and measurement method.	Completeness and confidence indicators.
3. Reliability scoring	Normalizes the five dimensions to 0-100 and computes the composite score using published weights.	Dimension scores and overall IRS.
4. Availability-reliability gap detector	Compares availability and service performance when their scales can be aligned.	Gap flags and evidence.
5. GIS map and village profile	Shows village categories, component scores, evidence and nearby services.	Spatial overview and drill-down.
6. Priority engine	Combines deficit, population impact, criticality, vulnerability and lack of alternatives.	Ranked investigation/intervention list.
7. What-if intervention simulator	Changes an input or uplift assumption and recomputes score/priority.	Scenario comparison and cost-effectiveness.
8. Feedback and verification	Captures reports or field verification with timestamps and review status.	Updated evidence queue, not an unverified truth feed.

7.2 End-to-end workflow

STEP 1	COLLECT
STEP 2	NORMALIZE
STEP 3	SCORE
STEP 4	COMPARE
STEP 5	PRIORITIZE
STEP 6	SIMULATE
STEP 7	VERIFY

Permitted datasets / field observations -> shared village ID and geometry -> normalized dimension scores -> reliability gaps -> impact-aware priority ranking -> alternative intervention scenarios -> human review and post-intervention measurement.

7.3 Main screens

- **Command Centre:** average reliability, high-gap villages, data coverage, top priorities and last-refresh dates.
- **Interactive GIS Map:** village markers/choropleth coloured by reliability, filters for dimension and date, and a clear legend.
- **Village Deep Dive:** component scores, availability status, observed performance, gap, population, nearest services, confidence and sources.
- **Priority Workbench:** sortable list with reason codes, affected population, criticality and lack of alternatives.
- **What-if Simulator:** compare road repair, connectivity improvement, outage reduction or an additional public service point.
- **Data Quality View:** show stale, missing, low-sample or conflicting evidence rather than hiding it.

8. Infrastructure Dimensions and Suggested Indicators

The initial five-dimension design carries forward the uploaded concept. The indicators below are candidate measures; the pilot must choose only the metrics for which evidence is available and sufficiently comparable.

Dimension	Candidate indicators	Interpretation / caution
Digital connectivity (25%)	Broadband/mobility availability; test download/upload speed; latency; jitter; packet loss; percentage of successful tests; uptime if provided.	Availability and performance are separate. A few speed tests are snapshots, not a statistically representative annual uptime measure. TRAI MySpeed documents speed, latency, jitter, packet loss and geospatial performance functions. [9]
Electricity reliability (20%)	Outage count; outage duration; supplied hours; planned/unplanned outages; backup power at critical facilities.	Use authorized utility records, published indicators or verified local logs. Do not invent hourly outage history from a simple connection flag.
Road accessibility/reliability (20%)	Road condition; all-season passability; closure days; travel time; alternative routes; road passability; proximity to service locations.	Road presence is not all-season passability. The Rural Access Index provides a recognized 2 km all-season-road measure. [10][11]
Public digital infrastructure (15%)	CSC/telecentre location; open hours; service availability; power/network readiness; verified service interruption reports.	A facility point means it exists, not that it is open or able to complete a transaction at every visit.
Essential-service accessibility (20%)	Road-network travel time to health facilities, schools, markets, Panchayat, banking and emergency services; seasonal detours.	Use network travel time where possible, not straight-line distance only. Access is a proxy for opportunity, not a direct measure of quality/outcomes.

Optional later modules include water-supply reliability, sanitation, public transport, weather-related disruption and facility-level equipment uptime. Keep the 24-hour MVP tightly scoped so that every displayed score has a defined calculation and source.

9. Scoring Methodology and Explainability

A transparent baseline is preferable for the hackathon because it can be inspected, challenged and explained. Machine learning can be added later only where adequate historic, labelled data exists.

9.1 Component scores

Convert each indicator to a 0-100 scale using documented rules and thresholds. For good-direction indicators, higher means better (e.g., uptime). For bad-direction indicators, reverse the scale (e.g., outage duration, closure days, travel time). Thresholds must be stated in the README and applied consistently.

Example only: a normalized digital connectivity score may combine uptime, test success and speed/latency bands. A road score may combine passability, condition and closure history. The exact transformations must be documented and validated with domain guidance, not selected solely to produce attractive demo results.

9.2 Infrastructure Reliability Score (IRS)

The proposed prototype weights retain the five categories from the original concept. They are project-specific design choices, not an official government index.

```
IRS = 0.25D + 0.20E + 0.20R + 0.15P + 0.20S
```

Where D = digital connectivity reliability; E = electricity reliability; R = road accessibility/reliability; P = public digital infrastructure; S = essential-service accessibility. All component scores use 0-100, where 100 means the chosen service-performance standard is met.

Missing data must not be silently converted to zero. The system should calculate a coverage-adjusted score only if a minimum amount of weighted data is present, renormalize over observed dimensions, and prominently show which dimensions are missing. For example, if only digital and road data are available, the display must say “partial score” and show confidence rather than comparing it as fully equivalent to a complete profile.

9.3 Example categories

IRS score	Prototype label	Meaning
80-100	Highly reliable	Most measured dimensions meet the chosen prototype standard.
60-79	Adequate / watch	Some gaps need monitoring or targeted action.
40-59	Vulnerable	Material deficiencies affect dependable access.
0-39	Critical	Severe measured deficits; verify urgently before planning action.

These are provisional interpretive bands for a hackathon prototype, not an official government classification or a validated policy threshold.

9.4 Availability-reliability gap

When both can be expressed on comparable 0-100 scales:

```
Gap = Availability Score - Reliability Score
```

Illustrative example: a village has an availability score of 95 for broadband but a measured reliability score of 52, producing a 43-point gap. A high gap is a signal to investigate; it is not proof that one particular operator or department caused the problem.

Suggested demo flag: 20-34 points = notable; 35+ points = high gap. These trigger levels are configurable and should be validated in a pilot. A dimension without comparable measurement should display “gap unavailable” rather than manufacturing a number.

9.5 Confidence and data freshness

Every score should show at least: source, observation date, geographic coverage, sample count (where relevant), method, missing fields and confidence. A confidence score can be constructed as a transparent quality indicator from recency, coverage and source/method quality; it must not be confused with a statistical probability unless calibrated as one.

10. Priority Ranking and Intervention Simulator

10.1 Priority score

A low IRS is not the only reason to prioritize a village. A moderate deficit affecting a hospital or a large, isolated population may be more urgent than a lower score with multiple alternatives. InfraPulse therefore proposes a second index for intervention priority.

```
Priority = 0.35G + 0.25A + 0.20C + 0.10V + 0.10N
```

G = service deficit/gap severity; A = estimated people affected; C = criticality of the service; V = vulnerability of affected population; N = lack of nearby alternatives. Each term is normalized 0-100. Weights are provisional and must be visible to the user.

Prioritization must remain explainable. The interface should show reason codes such as “low all-season road passability”, “high population affected”, “nearest health facility has long network travel time”, or “no practical alternative route”.

10.2 Intervention simulator

For a selected village, the user can change candidate inputs and compare outcomes. Examples include improving a road-condition score, reducing estimated closure days, increasing internet uptime, reducing outage duration or adding an operational digital access point. The scoring engine recalculates IRS and priority, showing the component that changed and the assumptions used.

10.3 Compare impact and cost

Where reasonable cost estimates exist, compare candidate intervention value as:

```
Scenario value = Estimated IRS improvement / Estimated intervention cost
```

For the MVP, costs can be entered as explicit demo assumptions. The interface must label the output as an estimate, not a government-approved budget proposal or a causal impact forecast. Before field use, cost assumptions and expected service improvements require engineering and departmental validation.

10.4 Demonstration example (simulated)

Option	Illustrative IRS before	Illustrative IRS after	Score change	Interpretation
Road improvement	38	52	+14	Largest modeled improvement in this scenario.
Broadband improvement	38	44	+6	Improves digital service but leaves road gap.
Power reliability improvement	38	47	+9	Useful cross-sector benefit if power supports other facilities.

All values in this table are hypothetical to illustrate the simulator. They are not measurements from a named Indian village and must be labelled "simulated data" in the demo.

11. Data Strategy, Quality and Provenance

11.1 Candidate sources

Data need	Potential source / method	Use and caveat
Village identity, boundaries, population	Local Government Directory, Census-related sources, state/official tables where accessible.	Use stable official identifiers and record the source vintage. Boundary changes can invalidate joins.
Infrastructure baseline and services	Mission Antyodaya, Gram Manchitra, official departmental datasets. [2][3][4]	Use availability indicators as a baseline; distinguish asset existence from performance.
Roads and facility locations	PMGSY/GeoSadak open data, NRIDA GIS references, OpenStreetMap with required attribution. [5][6][16]	Check date, geographic completeness, license and actual condition. Road presence is not passability.
Rural broadband deployment	BharatNet and DoT/PIB programme status. [7][8]	Deployment/service-ready records are not a substitute for user-experience tests.
Telecom performance	TRAI MySpeed measures or authorized/usable public results; direct team test data if consented. [9]	Do not assume a public bulk API is available. Protect location privacy and avoid overgeneralizing small samples.
Electricity reliability	Utility/ DISCOM records, published performance reports, authorized outage logs, field-verified local observations.	Access and data format may vary by state; without observed data, display missing/low-confidence rather than fabricate.
Population exposed / essential services	Population tables, official health/education facility directories, geospatial points.	Use the best available geography; facility quality and opening status need separate verification.
Citizen / field reports	A form with village, time, category, optional photo and contact-free mode.	Reports are leads, not verified facts. Add duplicate detection, moderation and status fields.

11.2 Data contract

Every observation should include: record_id, village_id, infrastructure_dimension, indicator_name, value, unit, direction (higher/lower is better), observation_start, observation_end, geometry/location, data_source, retrieval_date, methodology, sample_count, verification_status, confidence, license/usage notes and whether the value is measured, administrative, inferred or simulated.

11.3 Data-quality checks

- Validate ranges and units (for example, uptime cannot exceed 100%, and travel time cannot be negative).
- Detect duplicate village names and use official identifiers where possible.
- Flag stale observations and conflicting sources instead of silently choosing one.
- Keep a field-level provenance trail so a score can be explained to a judge or planner.
- Do not compare indicators gathered over different time periods as if they were simultaneous.
- For small samples, show sample count and uncertainty; do not present a handful of speed tests as annual availability.

12. Technical Architecture and Implementation

12.1 Architecture

Layer	Responsibilities	Suggested implementation
Data connectors	Load CSV/GeoJSON and permitted service exports; later add approved APIs.	Python import jobs
Geospatial normalization	Canonical village IDs, boundary joins, coordinate validation and geometry.	GeoPandas; PostGIS; GeoJSON
Quality/provenance	Age, missingness, source, confidence and validation rules.	Pandas + validation functions
Scoring engine	Normalize measures, compute component scores, IRS and gaps.	Python functions; deterministic tests
Priority/simulator	Rank gaps and compare the effects of scenario inputs.	Transparent weighted model
API	Serve villages, profiles, scores, rankings and scenarios.	FastAPI REST endpoints
Frontend	Dashboard, map, filters, profile, ranking and simulator.	React + Vite + Tailwind
GIS rendering	Color-code villages/assets and show nearest-service context.	Leaflet or MapLibre
Storage	Store master village data, indicator observations, scores and audit events.	PostgreSQL + PostGIS; SQLite can support a smaller MVP
Deployment	Host app and API; protect secrets and log errors.	Vercel + Render or equivalent

12.2 Suggested API endpoints

Endpoint	Purpose
GET /api/v1/villages	List pilot villages, geometry, score and data-coverage metadata.
GET /api/v1/villages/{id}	Return a selected village profile and source-linked indicators.
GET /api/v1/map?dimension=road	Return map-ready values for the selected dimension.
GET /api/v1/priorities	Return ranked villages, score components and reason codes.
POST /api/v1/simulate	Accept proposed changes and return recalculated scores/scenario comparison.
GET /api/v1/data-quality	Return stale/missing/low-confidence observations.
POST /api/v1/reports	Submit a public/field report as unverified pending review.

12.3 Suggested data entities

Village: id, official code, name, district/block/mandal, centroid, boundary, population and data vintage.
Asset: asset id, type, coordinates, operator/department if public, operational status, source and update date.
Observation: village/asset id, indicator, value, unit, time window, sample count, method, source, verification and confidence.
ScoreSnapshot: component scores, IRS, availability index, gap, coverage, threshold version and calculated date.
Scenario: baseline snapshot, intervention type, changed assumptions, expected score change, estimated cost and scenario label.

13. 24-Hour Hackathon MVP

The team must implement a coherent decision loop rather than a large catalogue of unfinished features. The minimum viable prototype is a map, score, gap, explanation, priority list and working what-if simulation, all driven by the same data model.

Feature	Status	Acceptance test
Interactive village GIS map	Must-have	A user can select a village and filter by overall/dimension score.
Village profile	Must-have	Shows each component, availability indicator, reliability measure, date and confidence.

Feature	Status	Acceptance test
IRS calculation	Must-have	Recalculates deterministically from displayed inputs and documented weights.
Gap detection	Must-have	Flags gap only when availability and reliability values are comparable.
Priority ranking	Must-have	Shows ordered villages and a human-readable explanation for each rank.
What-if simulator	Must-have	Changing one intervention changes the component score, IRS and ranking consistently.
Data quality view	Must-have	Shows missing, stale or simulated values clearly.
Field/citizen reports	Optional	A report enters as unverified and can be reviewed.
Live sensor / API integration	Post-hackathon	Only implement if source access and data permission are confirmed.

13.1 Recommended demo geography and data strategy

Select one compact geography with enough mapped villages to make the GIS meaningful. Prefer a pilot where public road/facility data and at least some connectivity or service observations can be obtained. If high-frequency data is unavailable, create a small simulated dataset for demonstrating interactions, mark every simulated value visibly, and avoid presenting it as real research results.

13.2 Example village dataset fields

```
village_id, village_name, district, latitude, longitude, population, internet_available,
internet_score, internet_sample_count, power_connection, power_reliability_score,
outage_observation_period, road_score, closure_days, hospital_travel_time, school_travel_time,
csc_available, csc_operational_score, availability_index, IRS, coverage_ratio,
confidence_level, source_updated_at.
```

13.3 24-hour execution schedule

Elapsed time	Deliverable
0-2 hours	Freeze exact MVP, user story, data schema, pilot geography and demo narrative.
2-5 hours	Acquire/prepare permitted base data; generate a clearly labelled synthetic extension only if needed.
5-8 hours	Implement backend data model, normalization, score and data-quality tests.
8-12 hours	Build GIS map, filters and village profile.
12-15 hours	Implement gap detector, priority ranking and reason codes.
15-18 hours	Implement what-if simulator and side-by-side scenario results.
18-20 hours	Integrate API/frontend, seed demo data and deploy.
20-22 hours	Test edge cases, score reproducibility, map rendering and missing-data behavior.
22-23 hours	Prepare slides, source/data explanation, README and backup screenshots/video.
23-24 hours	Freeze features; run final demo path and fix only critical defects.

14. Validation, Evaluation and Impact Measurement

InfraPulse should be judged not by the number of charts but by whether its rankings and explanations are transparent, reproducible and useful. A hackathon prototype can demonstrate method and interaction; claims of real-world predictive accuracy require a representative dataset and validation.

Evaluation area	Prototype test	Metric / evidence
Data correctness	Compare a sample of imported values with their source records.	Source-match rate; invalid-value count.
Scoring reproducibility	Recalculate the same village record multiple times.	Identical output for identical inputs and model version.
Missing data	Remove a component and check the result.	Missing dimensions flagged; no silent conversion to zero.

Evaluation area	Prototype test	Metric / evidence
Temporal quality	Use old and fresh observations.	Age and stale-data indicators visible in the UI.
Gap logic	Test cases with equal and disparate availability/performance values.	Correct gap only when scales are comparable.
Priority explanation	Inspect top-ranked villages.	Every ranking has reason codes and visible weighted factors.
Scenario consistency	Change one input and recalculate.	Scores change according to documented model; baseline retained.
Field validity	Compare selected map/profile with field checks or knowledgeable local review.	Documented agreement/disagreement; revise thresholds where warranted.
Decision usefulness	Ask a small group of intended users to choose a priority from the same cases.	Clarity and usefulness feedback; not a causal impact claim.

14.1 Possible pilot KPIs

- Percentage of pilot villages with adequate input coverage for a complete IRS.
- Number and share of villages with high availability-reliability gaps in each dimension.
- Number of residents plausibly affected by high-priority deficits, with population source and estimate date shown.
- Median road-network travel time to selected essential services, compared before/after a proposed scenario.
- Percentage of priority recommendations reviewed and accepted, amended or rejected by a domain expert.
- Observed change in the selected performance indicators after an intervention, where post-intervention data is available.

Expected outcomes must be expressed as prototype capabilities until verified. “Villages assessed”, “gaps flagged” and “scenario score change” are measurable prototype outputs. “Reduced travel time”, “improved access” or “increased reliability” are real-world outcomes that require implementation and follow-up measurement.

15. Risks, Constraints and Responsible Use

Risk	Why it matters	Control
Sparse performance data	A village may look reliable simply because no one measured its service.	Show coverage/confidence; never treat missing as zero or as success.
Stale records	Infrastructure and road conditions change.	Show observation date; configure expiry windows and stale flags.
Sampling bias	People who can run a speed test are not necessarily representative of every household.	Show sample count and location/time; avoid claiming village-wide representativeness from few tests.
False precision	A score like 63.27 can look more accurate than the evidence allows.	Display rounded score, ranges/labels and component evidence; explain it is an index.
Incomparable metrics	Availability, uptime and access distance measure different things.	Normalize with documented rules and keep components visible.
Ranking harms / stigma	A red village label may unfairly imply community failure.	Use “priority for verification/intervention”; explain causes and uncertainty, not blame.
Sensitive information	Detailed household locations or reports may expose residents.	Aggregate where possible; collect minimal personal data; consent, access controls and retention limits.
Incorrect recommendations	A simulator may ignore engineering constraints, budget rules or local context.	Label scenarios as indicative; require expert and community review before action.
Model bias	Weights may systematically favor well-measured areas or a selected service type.	Publish weights, test sensitivity, audit geographic coverage, and allow policy review.
Overclaiming AI	Synthetic demo outputs can be mistaken for validated predictions.	Mark simulated data, list model assumptions and report validation status honestly.

15.1 Safe claims

- **Acceptable:** “The prototype uses a transparent, configurable scoring model to surface reliability gaps using the provided dataset.”
- **Acceptable:** “The scenario simulator estimates how component-score changes affect the overall score under stated assumptions.”
- **Avoid unless validated:** “Our AI predicts actual failures with 95% accuracy,” “we have real-time government integrations,” or “this intervention will reduce travel time by a measured amount.”

16. Roadmap and Scaling

Stage	Scope	Exit criteria
Hackathon MVP	One pilot geography; five dimensions where data allows; scoring, gap, map, ranking and simulator.	End-to-end demo; traceable dataset; tests for scores and missing data.
Pilot validation	Work with a local authority/NGO/utility or field partner; verify selected indicators.	Documented data agreement, field checks, calibrated indicators and user feedback.
Operational prototype	Automated refresh for permitted datasets; access controls; audit trail; issue workflows.	Reliable refresh, data-quality monitoring and documented ownership.
Multi-district expansion	Reusable connectors, configurable weights, regional comparisons and budget constraints.	Comparable methodology plus local calibration and governance.
Long-term platform	Historical trends, early-warning signals, resilience analysis and scenario evaluation.	Validated time series, reliable integrations and post-intervention outcome measurement.

16.1 Scalability principle

Scale the data pipeline and governance before scaling the AI. Additional districts are useful only if village identity, data definitions, missing-data rules and score interpretation remain comparable. Regional weights can be configurable, but changes must be versioned so a village’s score does not appear to change for unexplained reasons.

17. Submission-Ready Problem Statement and Proposed Solution

17.1 Problem statement

Rural infrastructure is commonly assessed by whether an asset or service is recorded as available. However, availability alone does not show whether residents can reliably use it. Villages may have roads, electricity, broadband and public digital infrastructure on paper while experiencing outages, poor network performance, seasonal road inaccessibility, or excessive travel time to essential services. The evidence needed to diagnose these issues is distributed across surveys, geospatial databases, programme dashboards, utility records and service-performance measurements with different geographic units, definitions and update cycles. This makes it difficult to compare villages fairly, identify where nominal coverage masks poor service, estimate who is affected, and justify which intervention should come first.

17.2 Proposed solution

We propose **InfraPulse: Rural Infrastructure Reliability Intelligence Platform**, a GIS-based decision-support layer that combines infrastructure availability with measured or clearly labelled proxy indicators of reliability and accessibility. It builds a village-level infrastructure profile across digital connectivity, electricity, roads, public digital infrastructure and access to essential services. A transparent and configurable Infrastructure Reliability Score summarizes the indicators while keeping the component values, data dates, missingness and confidence visible. InfraPulse detects availability-versus-reliability gaps, maps underserved villages, explains which factors contribute to low scores, ranks potential interventions by service deficit, population affected, criticality, vulnerability and lack of alternatives, and simulates the expected score changes under candidate improvements.

The platform is designed to complement, not replace, existing systems such as Mission Antyodaya, Gram Manchitra, PMGSY/GeoSadak, BharatNet and TRAI’s network-measurement tools. It reuses public or authorized data where possible and clearly labels simulated values where field-level performance measurements are not available. Its initial goal is not to claim validated forecasting; it is to make rural infrastructure reliability gaps more visible, comparable and actionable, and to create a path toward field-validated planning support.

17.3 Innovation statement

The proposed innovation is the combination of a cross-sector availability-reliability gap, data confidence and freshness, population/criticality-aware ranking, and an interactive intervention simulator in one village-level workflow. GIS, asset mapping, rural infrastructure indicators, network-quality tests and infrastructure digital twins already exist individually; InfraPulse differentiates by tying reliable service delivery to explainable intervention choices.

17.4 One-line pitch

“InfraPulse reveals the gap between infrastructure that exists on paper and infrastructure that reliably serves rural communities - then helps decision-makers choose what to fix first.”

17.5 60-second pitch outline

- **Problem:** An asset marked as available may still fail residents through poor service quality, outages, seasonal closures or limited access.
- **Evidence:** India already has village surveys, GIS assets, rural road data and sector tools, but those sources have different purposes and data structures. [2]-[11]
- **Solution:** InfraPulse joins available and performance evidence, creates a transparent reliability profile, and displays where availability masks poor service.
- **Decision value:** It explains who is affected, ranks what needs attention first and lets users compare intervention scenarios.
- **Honesty:** The prototype labels simulated data, reports confidence and does not claim unvalidated predictive accuracy.
- **Impact:** Better evidence for more equitable, efficient and reliable rural infrastructure planning.

18. Research References and Existing-System Links

Public-source review accessed on 9 October 2026. Links are included so judges and teammates can inspect the original source. Descriptions below summarize the portions used in this document; they do not imply endorsement by the source organizations.

[1] United Nations DESA. *Goal 9: Industry, Innovation and Infrastructure.* Target 9.1 explicitly calls for quality, reliable, sustainable and resilient infrastructure with affordable and equitable access. Open source (accessed 2026-10-09).

[2] Ministry of Rural Development / NIC. *Mission Antyodaya 2022-23 - overview and survey.* Describes village/Gram Panchayat data, multi-sector indicators, ranking, composite score and gap reports for planning. Open source (accessed 2026-10-09).

[3] Ministry of Rural Development / NIC. *Mission Antyodaya FAQ.* Describes the 2022-23 village-level survey and its 216 indicators / 21 sections, including infrastructure and services. Open source (accessed 2026-10-09).

[4] Ministry of Panchayati Raj / NIC. *Gram Manchitra.* Shows Panchayat profiles, education, health, CSCs, PDS, roads, streetlights, asset reporting and proximity/service-area tools. Open source (accessed 2026-10-09).

[5] Press Information Bureau, Ministry of Rural Development. *Rural Connectivity GIS Data Released in Public Domain (22 February 2022).* Reports public release of GIS data covering 800,000+ rural facilities, 1 million+ habitations and 2.5 million km of rural roads; mentions open data and related systems. Open source (accessed 2026-10-09).

[6] National Rural Infrastructure Development Agency. *Database and Masterplan for Rural Roads.* Describes GIS layers, spatial/attribute/operational data, roads, terrain, census, facilities and traffic data for rural road planning. Open source (accessed 2026-10-09).

[7] Press Information Bureau, Ministry of Electronics and IT. *Government expands digital infrastructure; BharatNet and CSC status (6 August 2026).* Reports about 2.21 lakh Gram Panchayats service-ready under BharatNet as of June 2026 and 3.91 lakh rural Gram Panchayat-level CSCs as of May 2026. Open source (accessed 2026-10-09).

[8] BharatNet / Department of Telecommunications. *Amended BharatNet Programme.* Describes rural broadband/FTTH connectivity and service offerings; useful as an availability and service-enabling source. Open source (accessed 2026-10-09).

[9] Telecom Regulatory Authority of India. *TRAI revamps MySpeed App to check real-time network performance (13 February 2026).* Lists download/upload speed, latency, jitter, packet loss, continuous/scheduled testing and location-based historical performance maps. Open source (accessed 2026-10-09).

[10] World Bank. *Rural Access Index (RAI) - Data Catalog.* Defines RAI as rural population within 2 km of an all-season road; describes it as SDG indicator 9.1.1 and lists temporal/granularity metadata. Open source (accessed 2026-10-09).

[11] World Bank SDG Metadata. *Indicator 9.1.1 - Rural Access Index.* Explains geospatial inputs, road passability, population distribution, and data quality/update limitations for the indicator. Open source (accessed 2026-10-09).

[12] ACCESSLab / GitHub. *Rural Infrastructure Assessment.* Public project README describes route-level readiness scores combining lane markings, GPS quality, mobile broadband and HD-map availability, with maps and dashboards. Open source (accessed 2026-10-09).

[13] Devfolio / Hack-a-Sol 2024. *Hills to Highways.* Hackathon project describing GIS/ML-based rural and hilly road planning using population, traffic, geography and economic factors; listed as a track winner. Open source (accessed 2026-10-09).

[14] GitHub project repository. *Village Digital Twin / RuraLens.* Public project repository describes a rural digital twin, maps and a graph-based infrastructure failure/cascade simulator. This is treated as a project prototype, not a verified public-sector deployment. Open source (accessed 2026-10-09).

[15] GitHub project repository. *SmartGram - Rural Village Digital Twin.* Public project repository describes GIS, village infrastructure dashboards and population/resource forecasting. This is treated as a community prototype. Open source (accessed 2026-10-09).

[16] GeoSadak / PMGSY. *Rural Connectivity GIS Data - Open Data.* Source location for PMGSY rural-connectivity datasets where available and permitted for reuse. Open source (accessed 2026-10-09).

Appendix A. Minimum evidence required for a credible demo

- A compact pilot boundary with official village IDs or a clearly documented village list.
- At least one availability indicator and one service-performance or access indicator per dimension being scored.
- Visible source, date, unit, data status (measured, administrative, inferred or simulated) and confidence for each key value.
- Deterministic scoring and at least a small set of automated tests for edge cases and missing values.
- A demo scenario in which a documented input change causes a predictable, traceable change in the score and ranking.
- A slide or README statement that sample values and weights are prototype assumptions unless validated.

Appendix B. Terminology for the demo

Service-ready: a programme status meaning infrastructure is treated as ready under that programme's definition; do not equate it automatically with continuous end-user reliability.
Measured performance: observation from a defined test or operational record with time and method.
Proxy: an indirect indicator used when the ideal measure is unavailable; label it and explain the limitation.
Simulated: a value generated for demonstration; never present it as measured real-world evidence.
Reliability index: a designed composite score with explicit weights and thresholds, not an official government rating.

Closing principle: The purpose of InfraPulse is not to create another map of rural assets. It is to help make the reliability of services visible, show the evidence behind each gap, and support better choices about what to improve first.